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ORIGINAL PAPER

Isolated serum hepatitis B core antibody seropositivity: A diagnostic risk factor for occult hepatitis B infection

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Abstract

Anti-HBc screening and nucleic acid testing for hepatitis B viral DNA (HBV DNA) detection in blood donors are not routinely performed in clinical settings in Nigeria. This raises serious concerns for safety of blood at a time that global health standards advocate for transfusion safety. The aim of this research is to investigate if presence of anti-HBc in blood donors is actually associated with occult hepatitis B infection through basic and advanced procedures. Prospective blood donors who were seronegative for HBsAg but sero-positive for anti-HBc (with or without other markers) among the four hundred and seventy enrolled in a cross-sectional study were selected for this study. Samples were further screened for hepatitis B core immunoglobulin M by enzyme-linked immunosorbent assay. Nucleic acid testing was performed for confirmation of occult hepatitis B infection. Anti-HBc was detected in 20 (32.8%) of the sixty-one HBsAg antigen-negative blood donors which constituted 13.0% of the total number of enrolled blood donors. Anti-HBc total-positive differentiation showed eighteen (90.0%) anti-HBc (IgG) and two (10%) anti-HBc (IgM) were detected. Nucleic acid testing showed 5.0% prevalence of occult hepatitis B infection among the anti-HBc-positive blood donors with estimated HBV DNA viral load of 58 IU/ml. Demonstration of 5.0% occult hepatitis B prevalence showed the possibility of post-transfusion hepatitis B infection in transfusion recipients with consequent possible liver damage should hepatitis B surface antigen alone be the continued practice in clinical settings. The inclusion of antibody to hepatitis B core antigen screening in addition to hepatitis B surface antigen marker is an essential step to ensuring optimal blood safety and prevent post-transfusion hepatitis.

Key words: anti-HBc, isolated, Occult hepatitis B, risk factor, seropositivity.

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Introduction

Though blood transfusion, like tissue or organ transplant, offers meaningful benefits in several clinical situations, yet it is associated with some risks which include occult

hepatitis B infection (OBI). Occult hepatitis B infection is one of the major concerns in blood transfusion practice [1]. The risk becomes greater in facilities with limited resources such as those in developing nations bedevilled with high endemicity of hepatitis B virus (HBV). Blood donors with high-titre antibodies to hepatitis B core antigen (anti-HBc) have the potential of transmitting OBI to recipients of blood transfusion with possible consequent post-transfusion hepatitis B. [2,3]. Thus stringent screening for this viral marker as recommended by the World Health Organization becomes imperative [4]. Anti-HBc has been established as an independent biomarker of moderate to severe hepatic inflammation [5]. Experts have defined OBI as the presence of circulating HBV DNA in the absence of hepatitis B surface antigen (HBsAg), excluding the window period [6] and it is characterized by low and fluctuating viral load (<100 IU/ml) which often results in intermittent DNA detection [7]. In primary OBI, there is detectable HBV DNA without any detectable serological marker, be it anti-HBc or antibody to HBsAg (i.e. anti-HBs) [8]. Blood that is collected during the early window period of HBV infection is highly infectious [7]. Prevalence of anti-HBc has been reported by several authors especially in HBsAg-negative blood donors [8–10] thus presenting a suspected risk of HBV transmission through blood and blood products. In Nigerian Health Institutions, up to date, there is no established policy exemplifying the importance of screening for anti-HBc in addition to HBsAg in blood donors. Nucleic acid testing (NAT) is not done routinely either. We hypothesized that certain HBsAg-negative blood donors who might have been certified fit for donation might have detectable anti-HBc and HBV DNA which are important diagnostic indices for OBI [7,11–18]. Caviglia and his co-researchers recently corroborated the established fact that the presence of covalently closed circular DNA is strongly associated with anti-HBc seropositivity [19]. Studies on health implications of OBI have suggested that it is one of the factors responsible for HBV reactivation and an important risk factor for the development of hepatocellular carcinoma (HCC), the 6th leading cancer worldwide and the 5th and 9th most common cancer in men and women, respectively [10,20–23].

Strictly speaking, optimal safety of blood and reduction of HCC risk depend largely on stringent screening methods and timely diagnosis. There is a virtual lack of data on anti-HBc prevalence and confirmation of OBI in blood donors in Nigeria. The important questions are as follows: Will the inclusion of anti-HBc screening offer any benefits to safe blood practice? How infectious are anti-HBc positive donors? Due to high HBV endemicity in Nigeria, what possible strategies can be adopted to eliminate false anti-HBc-positivity, its associated increased discard rate and cost implications? Will current study on OBI provide any evidence for prompt policy formulation on the inclusion of anti-HBc screening in addition to routine HBsAg screening in blood donors? Is Nigeria ready for routine screening of blood with NAT? This study advanced HBV testing to include hepatitis B core immunoglobulin M or anti-HBc (IgM) determination by ELISA technique and HBV DNA detection by real-time polymerase chain reaction (qPCR) for optimal OBI confirmation. Other markers screened along with HBsAg and anti-HBc were simply part of the rapid one-step HBV multi-panel enzyme immunoassay kit and were only mentioned where necessary in this study.

Materials and methods

Study design

This study involved two distinct designs: Descriptive cross-sectional design which enabled consecutive selection of HBsAg-negative/anti-HBc-positive (with or without other markers) blood donors from a population of four hundred and seventy subjects who consented to participate in the study. Cohort anti-HBc-positive samples were further tested for anti-HBc (IgM) and HBV DNA to detect the 'window period' phase and confirm OBI, respectively. A structured questionnaire was used to obtain relevant demographic data. The age range of prospective blood donors enrolled was 16–60 years.

Exclusion criteria

Blood donors who: (1) declined participation (2) were diagnosed for HIV/AIDS (3) were HBsAg-positive or on treatment for hepatitis B infection as at the time of the study (4) were seronegative for all the viral markers on testing were excluded from the study.

Research methodology

Sample collection and storage

Four-millilitres of whole blood samples were collected into K_3EDTA sample container following informed consent. Plasma samples not analysed immediately were separated into plain containers for anti-HBc (IgM) and qPCR procedures were stored at −40°C till analysis.

Serologic assays

Hepatitis B viral markers (including anti-HBc) were screened for immediately using HBV 260 kit (Atlas Link, Inc., USA) for the initial selection of HBsAg-negative/anti-HBc-positive blood donors. Anti-HBc (IgM) determination by ELISA technique was performed to differentiate positive anti-HBc total into IgG and the IgM categories

using anti-HBc (IgM) kit (DIA.PRO diagnostic Bioprobes, Milano) according to the manufacturer instructions.

Nucleic acid testing assays

The nucleic acid testing assays for OBI confirmation were performed using real-time polymerase chain reaction or qPCR (Roche Molecular Systems, Pleasanton, CA) technique and HBV DNA 2.0 kit according to the manufacturer instructions.

Results

Out of 470 apparently, healthy blood donors enrolled, 61 (13.0%) were HBsAg-negative and 20 (4.3%) of these were anti-HBc-positive. Two (10.0%) of the anti-HBc-positive donors had detectable isolated anti-HBc (IgM) while remaining 18 (90.0%) were anti-HBc (IgG). The mean age of the participants in this study was 26.9 ± 7.2 years with almost 70% within the 21–40 years age group while around 20% were age 20 years or less. Male: female ratio was 1.2. Table 1 present the social demographic characteristics of research subjects and initial screening outcomes.

Anti-HBc prevalence, serologic differentiation into IgG and IgM and OBI confirmation

Anti-HBc$^+$ prevalence of 32.8% was observed among HBsAg-negative participants in this study. Eighteen (90.0%) of the twenty cases of anti-HBc$^+$ total were isolated anti-HBc-positive. One (5.0%) was in addition to antibody to hepatitis B surface antigen (anti-HBs$^+$) and antibody to hepatitis B envelope antigen (anti-HBe$^+$) markers and another one (5.0%) was detected in addition to HBeAg. Two (10.0%) of the isolated anti-HBc$^+$ samples were anti-HBc IgM-positive. None of the research subjects with isolated anti-HBc sero-positive cases was HBV DNA-positive. Overall, OBI was demonstrated in one (5.0%) of the overall anti-HBc sero-positive subjects investigated, specifically one with detectable anti-HBc and HBeAg as shown in Table 2.

Discussion

There was no clinically significant age difference between the overall blood donors enrolled and those who were HBsAg-negative. However, the mean age of overall blood donors enrolled was higher statistically. Up to 13.0% of the overall number of blood donors enrolled would have had their blood donations collected and stored in the blood bank based on HBsAg screening alone and that might be very risky for transfusion recipients as the analytical data later showed. The most common age group of both categories of research subjects enrolled was 21–40 years. Moreover, anti-HBc-positive donors were slightly higher among the 41–60 years age group compared to ≤20 years category. Cumulatively, from age 21–60 years, more than 75.0% of HBsAg-negative/anti-HBc positive donors engaged in blood donations. Based on study findings, particular attention should be paid to blood donors who are more than 20 years old while investigating anti-HBc seroprevalence. A number of studies that focused on gender difference among hepatitis B virus-infected subjects have been conducted [24,25]. This study outcome was not so different from previous findings by authors which indicated that HBV infections were more prevalent among male participants compared to their female counterparts [26,27]. We could not deduce from this study whether the outcomes of study have a connection with their employment status. Evidence showed that students participated more and anti-HBc seroprevalence was more among the students than other category of participants.

Differentiation of anti-HBc-positive is an essential step to enhancing the diagnostic classification of HBV infection into window phase and suspected occult hepatitis B infection. The only serologic evidence of hepatitis B infection in 2 (10.0%) of the 18 isolated anti-HBc$^+$ samples is the presence of anti-HBc (IgM) antibody. This was near twice the prevalence reported by Japhet *et al.* [28] but lower than that reported by Jeremiah *et al.* [29] who found 5.4% and 18.4% isolated HBcIgM seroprevalence among the blood donors in Osun State and in the semi-arid region of Nigeria, respectively. It has been proven to be the first serologic marker during the window phase of acute hepatitis B virus infection and usually disappears by the sixth month in those who recovered from the infection while anti-HBc (IgG) is an indicator of past or on-going infection and persists for a lifetime [30]. Moreover, the presence of acute hepatitis B infection during window period in which the only detectable marker is anti-HBc (IgM) has been published [31,32]. Authors have reported subsequent hepatocellular carcinoma in recipients of anti-HBc (IgM) during a nine-year follow-up study and recent international forum on OBI and transfusion safety corroborated this [1] with a look back donated sampling.

Occult hepatitis B infection has been defined as the presence and persistence of a circular, episomic form of HBV DNA (cccDNA) in the nucleus of infected cells in the absence of HBsAg in plasma (or serum) and with circulating HBV DNA < 200 IU/ml, with or without antibodies to HBsAg and hepatitis B core antigen (HBcAg) [6]. A more recent study put the associated circulating HBV DNA level at <100 IU/ml [7]. El-Ghitany *et al.* in a study reported that in OBI HBV DNA detection in serum or

Table 1 Social demographic characteristics of enrolled blood donors

Social Demog. Variable	Mean age ± SD	% Age difference	n (%)
Overall Blood donors enrolled:			470 (100.0)
Mean age (in Years)	26.9 ± 7.5		
HBsAg-negative/anti-HBc-positive		5.2	61 (13.0)
Donors:			
Average age (in Years)	25.7 ± 7.1		
Age groups (in Years)			
Overall Blood donors enrolled			
≤0			111 (23.6)
21–40			321 (68.3)
41–60			38 (8.1)
HBsAg-negative/anti-HBc-positive donors			
≤20			13 (21.3)
21–40			34 (55.7)
41–60			14 (23.0)
Sex:			
Overall Blood donors enrolled			
Male			278 (59.1)
Female			192 (40.9)
Male: Female ratio			1.5:1
HBsAg-negative/anti-HBc-positive Donors			
Male			11 (18.0)
Female			9 (14.8)
Male: Female ratio:			1.2:1
Employment status:			
Overall Donors enrolled			
Civil servants			124 (26.4)
Self-employed			81 (17.2)
Students			265 (56.4)
HBsAg-negative/anti-HBc-Positive:			
Civil servants			17 (27.9)
Self-employed			11 (18.0)
Students			33 (54.1)

anti-HBc, antibody to hepatitis B core antigen; HBsAg, Hepatitis B surface antigen.

hepatocytes of HBsAg-negative/anti-HBc (IgG)-positive donors may be with or without other markers [33]. Absence of HBV DNA in isolated anti-HBc-positive blood donors in this study would have implied 0% OBI prevalence based on anti-HBc screening alone. HBV DNA was detected only in blood donor's sample with unusual markers (anti-HBc and HBeAg) [8]. Thus 5.0% OBI prevalence from this study corroborated El-Ghitany's report and exemplifies the contribution of HBeAg, a marker of replication, in detecting OBI and predicting the risk for hepatocellular carcinoma. The molecular mechanisms that underlie OBI include: mutations in group specific 'a' determinant of the HBsAg which cause low binding affinity of HBsAg with monoclonal antibodies; mutations in the pre-S1 and pre-S2 regions, and RNA splicing steps which interfere with gene expressions and make HBsAg undetectable in blood donors by conventional assays [34]. Moreover, absence of HBV DNA in isolated anti-HBc does not necessarily imply the absence of OBI as there is the possibility of having HBV DNA level below the limit of detection.

While some authors have reported 0% prevalence of OBI among blood donors [35], and others reported 1.98%–2.7% [36,37], 5.0% OBI prevalence was found in this study. This was higher than the latter data reported. Approximately, the same finding was reported by Olotu *et al.* in a cohort study involving 354 (70.5%) of HBsAg-negative, anti-HBc sero-positive blood donors in Ile-Ife [11]. They reported 5.4% OBI prevalence. The lower population of HBsAg-negative and anti-HBc sero-positive research subjects enrolled for this study, it was thought, would create some bias but the outcome of this study ruled out the hypothesis. The evidence-based finding and the corroborating report of Olotu *et al.* implied that the current practice of transfusing blood based on HBsAg-negative blood alone still poses some threats to public health especially recipients of blood transfusion.

While the potential of HBV transmission through blood sero-positive for anti-HBc alone has been of great public health concern, whether screening for anti-HBc in addition to the HBsAg serologic marker for blood donation purpose is mandatory or not has been a matter of continuous debate especially in high endemic regions. The consequences of transfusion-acquired OBI on recipients such as cryptogenic liver disease, cirrhosis or hepatocellular carcinoma especially in chronic HBV cases warrant the routine screening. This position was supported by the World Health Organization's recommendation for both HBsAg and anti-HBc screening for blood transfusion purpose [38]. Studies showed that advances in screening protocol have led to a significant reduction in the HBV infection risk through blood transfusion. Reports showed that progressive reductions to 1 in 280 000 donations, 1 in 357 000 donations and 1 in 1 million donations were observed on the introduction of HBsAg, anti-HBc screening and mini pool nucleic acid testing for HBV DNA, respectively [39,40].

Three factors have cast doubts on the feasibility and sustainability of screening blood donations for anti-HBc in addition to HBsAg which include the relevant number of blood donations that will be lost (higher discard rate), the evidence that only a small fraction of anti-HBc-positive donors are infectious, especially those with detectable HBV DNA in plasma/serum [10,11,29,41] and the relatively high frequency of non-specific results for anti-HBc in the screening of a healthy population [42].

Table 2 Anti-HBc Prevalence, Serologic Differentiation into IgG and IgM and OBI Confirmation

HBsAg-Negative			
Research subjects: n (%)			**61 (100.0)**
Total anti-HBc$^+$ prevalence			**20 (32.8%)**
Cohort anti-HBc$^+$ Cases for OBI Confirmation: n (%)	**HBV DNA$^+$**	**HBV DNA (IU/ml)**	**HBV DNA**
20 (100.0)	1 (5.0)	58.0	19 (95.0)
Anti-HBc$^+$ Categorization for OBI Confirmation:			
Isolated anti-HBc$^+$ (IgG) prevalence	0 (0)	-	16 (80.0)
Anti-HBc$^+$ (IgM)	0 (0)	-	2 (10.0)
Anti-HBs$^+$, anti-HBe$^+$, anti-HBc$^+$ (IgG)	0 (0)	-	1 (5.0)
Anti-HBc$^+$(IgG), HBeAg$^+$	1 (5.0)	58.0	0 (0)

Prevalence of OBI in Anti-HBc$^+$ Donors: 5.0%.

OBI Prevalence among the overall population of donors screened for Blood Safety: 1 in 470 (Nearly 1 in 500).

Strategies for resolving the controversial decisions on anti-HBc inclusion in HBV screening protocol for blood transfusion/organ transplant purpose

Optimal donor selection methods

Solving the age-long controversies on anti-HBc inclusion or exclusion in HBV screening for blood transfusion/organ transplant requires multiple strategies. Firstly, optimal blood donor selection methods that promote 100% voluntary non-remunerated blood donation according to vision 2020 set by the World Health Organization [43,44] and reverse the current trend of 30%, 60% and 10% replacement, commercial and voluntary blood donations respectively [45] should be adopted in Nigeria.

Optimization of hepatitis B viral markers screening limit of detection

To obtain quality data, promote optimal blood safety and eliminate residual HBV risk, HBV kits to be selected must be able to detect HBsAg within a range of 0.04 ng/ml (or 0.01 IU/ml) [46,47] to 0.2 ng/ml (or 0.05 IU/ml) [48] of HBsAg and 2000 National Clinical Unit/L anti-HBc total in sera/plasma samples [49]. The US Food and Drug Administration suggested that HBsAg kit that fails to detect 0.5 ng/ml of HBsAg should not be used in screening blood [50].

Blood transfusion in selective category of patients

Owing to the association of anti-HBc with occult hepatitis B infection, careful screening of blood units meant for transfusion into certain category of patients becomes highly imperative. The US Centre for Disease Control and Prevention insists on post-vaccination testing for antibody to hepatitis B surface antigen (anti-HBs) in these patients to establish their immune protection [51]. When blood transfusion becomes necessary in unvaccinated patients with HIV/AIDS, hepatitis C virus (HCV) infection and stem cell disorders, as well as those on haemodialysis, steroids and others undergoing chemotherapy for myeloproliferative or lymphoproliferative disorders, screening of suitable blood units for anti-HBc in addition to HBsAg, becomes inevitable. Hepatitis B viral replication or reactivation is common in these categories of patients and can affect prognosis in the clinical management of the primary disorders [34,52].

Implementation of nucleic acid testing

For Nigeria to meet up with the current trends globally and achieve the millennium development goals that target eradication of vaccine-preventable diseases (including HBV infections) as one of its objectives, gradual implementation of NAT for the screening of donation samples is non-negotiable. First, stakeholders have to draft a policy in collaboration with the National Blood Transfusion Service; with NGOs as implementing partners on NAT for donated blood either using mini pools or individual testing approaches. Secondly, NAT implementation may require that it is first implemented in phases at geopolitical zones in the country before being institutionalized.

Testing of plasma bags

Testing of aliquot plasma samples from donated bags rather than index samples is very essential if issues relating to contamination, cross-reactivity and false anti-HBc seropositivity are to be eliminated [1].

Conclusion

Hepatitis B surface antigen screening alone in prospective blood donors is not sufficient to ensure optimal blood safety, eliminate OBI and consequent post-transfusion hepatitis B in blood transfusion or stem cell/organ transplant recipients. Concomitant detection of a marker of viral replication (HBeAg) is a potent risk factor for occult

hepatitis B infection in isolated anti-HBc-positive blood donors. Conference of experts including hepatologists, gastroenterologists, blood transfusion scientists, virologists, haematologists, internal medicine physicians other relevant researchers should identify all risk factors of OBI and formulate workable policies in Nigeria health system that will eliminate residual HBV infection risk. These must include implementation of universal anti-HBc screening in addition to HBsAg and strategic introduction of nucleic acid testing in no distant time.

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Conflict of interest

Authors declared no conflict of interest.

References

1 Seed CR, Allain JP, Lozano M, *et al.*: International forum on Occult hepatitis B infection and transfusion safety. *Vox Sang* 2019; **114**:397–406
2 Mosley JW, Stevens CE, Aach RD: Donor screening for antibody to hepatitis B core antigen and hepatitis B virus infection in transfusion recipients. *Transfusion* 1995; **35**:5–12
3 Stramer SL, Notari EP, Krysztof DE, *et al.*: Hepatitis B virus testing by Mini pool nucleic acid testing: does it improve blood safety? *Transfusion* 2013; **53**:2449
4 World Health Organization: Guidelines on estimation of residual risk of HIV, HBV or HCV infections via cellular blood components and plasma. WHO 2015.
5 Zhou J, Song L, Zhao H, *et al.*: Serum hepatitis B core antibody as a biomarker of hepatic inflammation in chronic hepatitis B patients with normal alanine aminotransferase. *Sci Rep* 2017; **7**:2747–2756
6 Raimondo G, Allain JP, Brunetto MR: Statements from the Taormina expert meeting on occult hepatitis B virus infection. *J Hepatol* 2008; **49**:652–657
7 Kiely P, Margaritis AR, Seed CR, *et al.*: Hepatitis B virus nucleic acid amplification testing of Australian blood donors highlights the complexity of confirming occult hepatitis B virus infection. *Transfusion* 2014; **54**:2084–2091
8 Hollinger FB: Hepatitis B virus infection and transfusion medicine: science and the occult. *Transfusion* 2008; **48**:1001–1026
9 Manzini P, Abate ML, Valpreda C, *et al.*: Evidence of acute primary occult hepatitis B virus infection in an Italian repeat blood donor. *Transfusion* 2009; **49**:757–764
10 Velati C, Romanò L, Fomiatti L: Impact of nucleic acid testing for hepatitis B virus, hepatitis C virus, and human immunodeficiency virus on the safety of blood supply in Italy: a 6-year survey. *Transfusion* 2008; **48**:2205–2213
11 Olotu AA, Oyelese AO, Salawu L, *et al.*: Occult Hepatitis B virus infection in previously screened, blood donors in Ile-Ife, Nigeria: implication for blood transfusion and stem cell transplantation. *Virol J* 2016; **13**:76
12 Hollinger FB, Sood G: Occult hepatitis B virus infection: a covert operation. *J Viral Hepat* 2010; **17**:1–15
13 Urbani S, Fagnoni F, Missale G, *et al.*: The role of anti-core antibody response in the detection of occult hepatitis B virus infection. *Clin Chem Lab Med* 2010; **48**:23–29
14 Bremer CM, Saniewski M, Wend UC, *et al.*: Transient occult hepatitis B virus infection in a blood donor with high viraemia. *Transfusion* 2009; **49**:1621–1629
15 Reesink HW, Engelfriet CP, Henn G, *et al.*: Occult hepatitis B infection in blood donors. *Vox Sang* 2008; **94**:153–166
16 Manzini P, Girotto M, Borsotti R, *et al.*: Italian blood donors with anti-HBc and occult hepatitis B virus infection. *Haematologica* 2007; **92**:1664–1670
17 Raimondo G, Pollicino T, Cacciola I, *et al.*: Occult hepatitis B virus infection. *J Hepatol* 2007; **46**:160–170
18 Liu CJ, Chen DS, Chen PJ: Epidemiology of HBV infection in Asian blood donors: emphasis on occult HBV infection and the role of NAT. *J Clin Virol* 2006; **36**(Suppl 1):S33–S44
19 Caviglia GP, Pellicano R, Sarraco GM, *et al.*: Hepatitis B core-related antigen: a serum biomarker for intrahepatic covalently closed circular DNA. *Clin Lab* 2018; **64**:411–412
20 Pollicino T, Raimondo G: Occult hepatitis B. *J Hepatol* 2014; **61**:688–689
21 Kumar R, Perez-Del-Pulgar S, Testoni B, *et al.*: Clinical relevance of the study of hepatitis B virus covalently closed circular DNA. *Liver Int* 2016; **36**(Suppl 1):72–77
22 Gao J, Xie L, Yang WS, *et al.*: Risk factors of hepatocellular carcinoma - current status and perspectives. *Asian Pac J Cancer Prev* 2012; **13**:743–753
23 World Cancer Research Fund International. Global statistics for the most common cancers. Available at: https://www.wcrf.org/dietandcancer-trends/worldwide-cancer-data (Accessed on 10th May 2019).
24 Shaz BH, James AB, Hillyer KL, *et al.*: Demographic patterns of blood donors and donations in a large metropolitan area. *J Natl Med Assoc* 2011; **103**(4):351–357
25 Farshadpour F, Taherkhani R, Tajbakhsh S, *et al.*: Prevalence and trends of transfusion-transmissible viral infections among blood donors in South of Iran: an eleven-year retrospective study. *PLoS ONE* 2016; **11**(6):e0157615
26 Li X, Zheng Y, Liau A, *et al.*: Hepatitis B virus infections and risk factors among the general population in Anhui Province, China: an epidemiological study. *BMC Public Health* 2012; **12**:272
27 Baig S: Gender disparity in infections of hepatitis B virus. *J Coll Physicians Surg Pak* 2009; **19**(9):598–600
28 Japhet MO, Adesina OA, Emmanuel D, *et al.*: Hepatitis B Core IgM (anti-HBcIgM) among hepatitis B Surface antigen (HBsAg) negative blood donors in Nigeria. *Virol J* 2011; **8**:513–518

29 Jeremiah ZA, Idris H, Ajayi BB, *et al.*: Isolated anti-IgM antibody among blood donors in the semi-arid region of Nigeria. *Hum Antibodies* 2011; **4**:77–82

30 Centers for Disease Control and Prevention: A comprehensive immunization strategy to eliminate transmission of hepatitis B virus infection in the United States: recommendations of the Advisory Committee on Immunization Practices (ACIP); Part 1: immunization of Infants, Children, and Adolescents. *MMWR Morb Mortal Wkly Rep* 2005; **54**(16):1–32

31 Whalley SA, Murray JM, Brown D, *et al.*: Kinetics of acute hepatitis B virus infection in humans. *J Exp Med* 2001; **193**:847–854

32 Ganem D, Prince AM: Hepatitis B virus infection-natural history and clinical consequences. *N Engl J Med* 2004; **350**:1118–1129

33 El-Ghitany EM, Farghaly AG: Serological pattern of hepatitis B virus among HBsAg negative blood donors in Alexandria, Egypt. *East Mediterr Health J* 2013; **19**(7):600–607

34 Makvandi M: Update on occult hepatitis B virus infection. *World J Gastroenterol* 2016; **22**(39):8720–8734

35 Wolff FH, Fuchs SC, Brandao AB: Absence of occult hepatitis B among blood donors in Southern Brazil. *Braz J Infect Dis* 2011; **15**(2):159–162

36 Moresco MN, Virgolino HA, deMorais MP, *et al.*: Occult hepatitis B virus infection among blood donors from the Brazilian Amazon: implications for transfusion policy. *Vox Sang* 2014; **107**(1):19–25

37 Arrae LC, Ximenes R, Andrieu JM, *et al.*: The biological meaning of anti-HBe positive result in blood donors: relation to HBV DNA and other serologic markers. *Rev Inst Med Trop Sao Paulo* 2003; **45**(3):137–140

38 World Health Organization: *Guidelines on assessing donor's suitability for blood donation*. Geneva: WHO, 2015.

39 Hennig H, Puchta I, Luhm J: Frequency and load of hepatitis B virus DNA in first-time blood donors with antibodies to hepatitis B core antigen. *Blood* 2002; **100**:2637–2641

40 Kleinman SH, Kuhns MC, Todd DS: Frequency of HBV DNA detection in US blood donors testing positive for the presence of anti-HBc: implications for transfusion transmission and donor screening. *Transfusion* 2003; **43**:696–704

41 Zuccaro O, Tosti ME, Mele A, *et al.*: Epidemiology of acute viral hepatitis in Italy: results of the surveillance through SEIEVA. *Rapporti ISTISAN* 2012; **4**:1–24

42 Larrubia JR: Occult hepatitis B virus infection: a complex entity with relevant clinical implications. *World J Gastroenterol* 2011; **17**(12):1529–1530

43 Maya ET, Buntugu KA, Pobee F, *et al.*: Rhesus negative woman transfused with rhesus positive blood: subsequent normal pregnancy without anti-D production. *Ghana Med J* 2015; **49**(1):60–63

44 World Health Organization. Voluntary unpaid blood donations must increase rapidly to meet 2020 goal. 2016. Available from: http://who.int/mediacentre/news/releases/2016/world- blood-donor-day/en/ [Accessed on 31st May, 2019].

45 Shrivastava SR, Shrivastava PS, Ramasamy J: Aiming to achieve 100% voluntary blood donation in all nations: global vision. *Ann Trop Med Public Health* 2017; **10**(4):783–784

46 Federal Ministry of Health (2010). Voluntary blood donation accounts for 10% of Nigeria's total collection. Available at: www.m.guardian.ng/features/science/voluntary-blood- donation-accounts-for-10-of-nigerias-total-collection/

47 Ghosh M, Nandi S, Dutta S, *et al.*: Detection of hepatitis B virus infection: a systematic review. *World J Gastroenterol* 2015; **7**(23):2482–2491

48 Kaminska A, Witkowska E, Winkler K, *et al.*: Detection of Hepatitis B virus antigen from human blood: SERS immunoassay in a microfluidic system. *Biosens Bioelectron* 2015; **66**:461–467

49 Deguchi M, Yamashita N, Kagita M, *et al.*: Quantitation of Hepatitis B surface antigen by an automated chemiluminescent microparticle immunoassay. *J Virol Methods* 2004; **115**:217–222

50 Food and Drug Administration. Guidance for Industry: Adequate and Appropriate Donor Screening Tests for Hepatitis B; Hepatitis B Surface Antigen (HBsAg) Assays Used to Test Donors of Whole Blood and Blood Components, Including Source Plasma and Source Leukocytes. Available from: https://www.fda.gov/media/124336/download (Accessed on 2nd June 2019).

51 Centers for Disease Control and Prevention. A comprehensive immunization strategy to eliminate transmission of hepatitis B virus infection in the United States: recommendations of the Advisory Committee on Immunization Practices (ACIP); Part 1: immunization of Infants, Children, and Adolescents. *MMWR Morb Mortal Wkly Rep* 2005;**54**:1–32.

52 Zou S, Stramer SL, Notari EP: Current incidence and residual risk of hepatitis B infection among blood donors in the United States. *Transfusion* 2009; **49**:1609